# Improving Language Understanding by Generative Pre-Training

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## Abstract

Natural language understanding comprises a wide range of diverse tasks such as textual entailment, question answering, semantic similarity assessment, and document classification. Although large unlabeled text corpora are abundant, labeled data for learning these specific tasks is scarce, making it challenging for discriminatively trained models to perform adequately. We demonstrate that large gains on these tasks can be realized by *generative pre-training* of a language model on a diverse corpus of unlabeled text, followed by *discriminative fine-tuning* on each specific task. In contrast to previous approaches, we make use of task-aware input transformations during fine-tuning to achieve effective transfer while requiring minimal changes to the model architecture. We demonstrate the effectiveness of our approach on a wide range of benchmarks for natural language understanding. Our general task-agnostic model outperforms discriminatively trained models that use architectures specifically crafted for each task, significantly improving upon the state of the art in 9 out of the 12 tasks studied. For instance, we achieve absolute improvements of 8.9% on commonsense reasoning (Stories Cloze Test), 5.7% on question answering (RACE), and 1.5% on textual entailment (MultiNLI).

## 1 Introduction

The ability to learn effectively from raw text is crucial to alleviating the dependence on supervised learning in natural language processing (NLP). Most deep learning methods require substantial amounts of manually labeled data, which restricts their applicability in many domains that suffer from a dearth of annotated resources [61]. In these situations, models that can leverage linguistic information from unlabeled data provide a valuable alternative to gathering more annotation, which can be time-consuming and expensive. Further, even in cases where considerable supervision is available, learning good representations in an unsupervised fashion can provide a significant performance boost. The most compelling evidence for this so far has been the extensive use of pre-trained word embeddings [10, 39, 42] to improve performance on a range of NLP tasks [8, 11, 26, 45].

Leveraging more than word-level information from unlabeled text, however, is challenging for two main reasons. First, it is unclear what type of optimization objectives are most effective at learning text representations that are useful for transfer. Recent research has looked at various objectives such as language modeling [44], machine translation [38], and discourse coherence [22], with each method outperforming the others on different tasks.[1] Second, there is no consensus on the most effective way to transfer these learned representations to the target task. Existing techniques involve a combination of making task-specific changes to the model architecture [43, 44], using intricate learning schemes [21] and adding auxiliary learning objectives [50]. These uncertainties have made it difficult to develop effective semi-supervised learning approaches for language processing.

[1] https://gluebenchmark.com/leaderboard

Preprint. Work in progress.

# 通过生成式预训练提升语言理解能力

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## 摘要

自然语言理解涵盖广泛且多样的任务，例如文本蕴含、问答、语义相似度评估和文档分类。尽管大规模无标注文本语料库十分丰富，但用于学习这些特定任务的标注数据却非常稀缺，这使得判别式训练模型难以充分表现。我们证明，通过在多样化的无标注文本语料库上对语言模型进行生成式预训练，再针对每个特定任务进行判别式微调，可以在这些任务上取得显著提升。与以往方法不同，我们在微调过程中采用任务感知的输入变换，从而在几乎不改变模型架构的前提下实现高效迁移。我们在多个自然语言理解基准测试中验证了该方法的有效性。我们提出的通用任务无关模型，其表现优于那些针对每个任务专门设计架构的判别式训练模型，并在所研究的 12 个任务中，有 9 个任务显著超越了当前最先进水平。 例如，我们在常识推理（Stories Cloze Test）上取得了 8.9%的绝对提升，在问答任务（RACE）上提升了 5.7%，在文本蕴含任务（MultiNLI）上提升了 1.5%。

## 1　引言

从原始文本中有效学习的能力，对于减轻自然语言处理（NLP）中对监督学习的依赖至关重要。大多数深度学习方法需要大量人工标注的数据，这限制了它们在许多缺乏标注资源的领域中的应用[61]。在这些情况下，能够利用未标注数据中的语言信息的模型，为收集更多耗时且昂贵的标注提供了有价值的替代方案。此外，即使在有大量监督信息可用的情况下，以无监督方式学习良好的表示也能带来显著的性能提升。迄今为止，最有力的证据是预训练词嵌入的广泛使用[10, 39, 42]，它提升了多项 NLP 任务的性能[8, 11, 26, 45]。

然而，利用无标注文本中超越词汇级别的信息存在两大挑战。首先，尚不明确何种优化目标能最有效地学习对迁移有用的文本表征。近期研究探索了语言建模[44]、机器翻译[38]及语篇连贯性[22]等多种目标，但不同方法在不同任务上各有优劣。其次，如何将这些习得的表征高效迁移至目标任务尚未达成共识。现有技术需结合多种手段：对模型架构进行任务特定调整[43,44]、采用复杂的学习方案[21]以及添加辅助学习目标[50]。这些不确定性使得为语言处理开发有效的半监督学习方法变得困难重重。

[1] https://gluebenchmark.com/leaderboard

预印本。工作正在进行中。

In this paper, we explore a semi-supervised approach for language understanding tasks using a combination of unsupervised pre-training and supervised fine-tuning. Our goal is to learn a universal representation that transfers with little adaptation to a wide range of tasks. We assume access to a large corpus of unlabeled text and several datasets with manually annotated training examples (target tasks). Our setup does not require these target tasks to be in the same domain as the unlabeled corpus. We employ a two-stage training procedure. First, we use a language modeling objective on the unlabeled data to learn the initial parameters of a neural network model. Subsequently, we adapt these parameters to a target task using the corresponding supervised objective.

For our model architecture, we use the *Transformer* [62], which has been shown to perform strongly on various tasks such as machine translation [62], document generation [34], and syntactic parsing [29]. This model choice provides us with a more structured memory for handling long-term dependencies in text, compared to alternatives like recurrent networks, resulting in robust transfer performance across diverse tasks. During transfer, we utilize task-specific input adaptations derived from traversal-style approaches [52], which process structured text input as a single contiguous sequence of tokens. As we demonstrate in our experiments, these adaptations enable us to fine-tune effectively with minimal changes to the architecture of the pre-trained model.

We evaluate our approach on four types of language understanding tasks – natural language inference, question answering, semantic similarity, and text classification. Our general task-agnostic model outperforms discriminatively trained models that employ architectures specifically crafted for each task, significantly improving upon the state of the art in 9 out of the 12 tasks studied. For instance, we achieve absolute improvements of 8.9% on commonsense reasoning (Stories Cloze Test) [40], 5.7% on question answering (RACE) [30], 1.5% on textual entailment (MultiNLI) [66] and 5.5% on the recently introduced GLUE multi-task benchmark [64]. We also analyzed zero-shot behaviors of the pre-trained model on four different settings and demonstrate that it acquires useful linguistic knowledge for downstream tasks.

## 2 Related Work

**Semi-supervised learning for NLP** Our work broadly falls under the category of semi-supervised learning for natural language. This paradigm has attracted significant interest, with applications to tasks like sequence labeling [24, 33, 57] or text classification [41, 70]. The earliest approaches used unlabeled data to compute word-level or phrase-level statistics, which were then used as features in a supervised model [33]. Over the last few years, researchers have demonstrated the benefits of using word embeddings [11, 39, 42], which are trained on unlabeled corpora, to improve performance on a variety of tasks [8, 11, 26, 45]. These approaches, however, mainly transfer word-level information, whereas we aim to capture higher-level semantics.

Recent approaches have investigated learning and utilizing more than word-level semantics from unlabeled data. Phrase-level or sentence-level embeddings, which can be trained using an unlabeled corpus, have been used to encode text into suitable vector representations for various target tasks [28, 32, 1, 36, 22, 12, 56, 31].

**Unsupervised pre-training** Unsupervised pre-training is a special case of semi-supervised learning where the goal is to find a good initialization point instead of modifying the supervised learning objective. Early works explored the use of the technique in image classification [20, 49, 63] and regression tasks [3]. Subsequent research [15] demonstrated that pre-training acts as a regularization scheme, enabling better generalization in deep neural networks. In recent work, the method has been used to help train deep neural networks on various tasks like image classification [69], speech recognition [68], entity disambiguation [17] and machine translation [48].

The closest line of work to ours involves pre-training a neural network using a language modeling objective and then fine-tuning it on a target task with supervision. Dai et al. [13] and Howard and Ruder [21] follow this method to improve text classification. However, although the pre-training phase helps capture some linguistic information, their usage of LSTM models restricts their prediction ability to a short range. In contrast, our choice of transformer networks allows us to capture longer-range linguistic structure, as demonstrated in our experiments. Further, we also demonstrate the effectiveness of our model on a wider range of tasks including natural language inference, paraphrase detection and story completion. Other approaches [43, 44, 38] use hidden representations from a

本文探讨了一种结合无监督预训练与有监督微调的半监督方法，用于解决语言理解任务。我们的目标是学习一种通用表征，使其能够通过少量调整迁移至多种任务。我们假设可获取大规模无标注文本语料库，以及若干带有手动标注训练样本的数据集（目标任务）。该设置不要求目标任务与无标注语料库处于相同领域。我们采用两阶段训练流程：首先，基于无标注数据通过语言建模目标学习神经网络模型的初始参数；随后，利用对应目标任务的有监督目标对这些参数进行适配。

在模型架构方面，我们采用了 Transformer[62]，该架构已在机器翻译[62]、文档生成[34]和句法分析[29]等多项任务中展现出卓越性能。相较于循环网络等替代方案，这种模型选择为我们提供了更具结构化的记忆机制，能够更好地处理文本中的长距离依赖关系，从而在不同任务间实现稳健的迁移性能。在迁移过程中，我们采用了基于遍历式方法[52]的任务特定输入适配策略，将结构化文本输入处理为单一连续的词元序列。正如实验所证实的，这些适配策略使我们能够在几乎不改变预训练模型架构的前提下进行高效微调。

我们在四种语言理解任务上评估了我们的方法——自然语言推理、问答、语义相似度和文本分类。我们这种通用的任务无关模型，其表现优于那些针对每个任务专门设计架构的判别式训练模型，并在所研究的 12 个任务中，有 9 个任务显著提升了当前最优水平。例如，我们在常识推理（Stories Cloze Test）[40]上取得了 8.9%的绝对提升，在问答（RACE）[30]上提升了 5.7%，在文本蕴含（MultiNLI）[66]上提升了 1.5%，以及在近期推出的 GLUE 多任务基准测试[64]上提升了 5.5%。我们还分析了预训练模型在四种不同设置下的零样本行为，并证明它获得了对下游任务有用的语言知识。

## 2　相关工作

自然语言处理中的半监督学习

我们的工作大致属于自然语言半监督学习的范畴。这一范式已引起广泛关注，并应用于序列标注[24, 33, 57]或文本分类[41, 70]等任务。早期方法利用未标注数据计算词级或短语级统计量，随后将其作为监督模型的特征[33]。近年来，研究人员已证明使用在未标注语料上训练的词嵌入[11, 39, 42]能够提升多种任务的性能[8, 11, 26, 45]。然而，这些方法主要传递词级信息，而我们旨在捕捉更高层次的语义。

近期研究探索了从未标注数据中学习并利用超越词级语义的方法。通过未标注语料训练的短语级或句子级嵌入，已被用于将文本编码为适合各类目标任务的向量表示[28, 32, 1, 36, 22, 12, 56, 31]。

无监督预训练是无监督预训练是半监督学习的一种特殊情况，其目标是寻找良好的初始化点而非修改监督学习目标。早期研究探索了该技术在图像分类[20,49,63]和回归任务[3]中的应用。后续研究[15]表明，预训练作为一种正则化方案，能够提升深度神经网络的泛化能力。在近期工作中，该方法已被用于辅助训练各类任务的深度神经网络，包括图像分类[69]、语音识别[68]、实体消歧[17]和机器翻译[48]。

与我们最接近的工作是使用语言建模目标预训练神经网络，然后在目标任务上进行有监督的微调。Dai 等人[13]以及 Howard 和 Ruder[21]采用这种方法来改进文本分类。然而，尽管预训练阶段有助于捕捉一些语言信息，但他们使用的 LSTM 模型限制了其预测能力，使其只能处理短距离依赖。相比之下，我们选择的 Transformer 网络能够捕捉更长距离的语言结构，这一点在我们的实验中得到了验证。此外，我们还证明了我们的模型在更广泛的任务上的有效性，包括自然语言推理、释义检测和故事补全。其他方法[43,44,38]在目标任务上训练监督模型时，会使用来自预训练语言模型或机器翻译模型的隐藏表示作为辅助特征。

pre-trained language or machine translation model as auxiliary features while training a supervised model on the target task. This involves a substantial amount of new parameters for each separate target task, whereas we require minimal changes to our model architecture during transfer.

**Auxiliary training objectives** Adding auxiliary unsupervised training objectives is an alternative form of semi-supervised learning. Early work by Collobert and Weston [10] used a wide variety of auxiliary NLP tasks such as POS tagging, chunking, named entity recognition, and language modeling to improve semantic role labeling. More recently, Rei [50] added an auxiliary language modeling objective to their target task objective and demonstrated performance gains on sequence labeling tasks. Our experiments also use an auxiliary objective, but as we show, unsupervised pre-training already learns several linguistic aspects relevant to target tasks.

# 3 Framework

Our training procedure consists of two stages. The first stage is learning a high-capacity language model on a large corpus of text. This is followed by a fine-tuning stage, where we adapt the model to a discriminative task with labeled data.

## 3.1 Unsupervised pre-training

Given an unsupervised corpus of tokens $\mathcal{U} = \{u_1, \ldots, u_n\}$, we use a standard language modeling objective to maximize the following likelihood:

$$L_1(\mathcal{U}) = \sum_i \log P(u_i | u_{i-k}, \ldots, u_{i-1}; \Theta) \tag{1}$$

where $k$ is the size of the context window, and the conditional probability $P$ is modeled using a neural network with parameters $\Theta$. These parameters are trained using stochastic gradient descent [51].

In our experiments, we use a multi-layer *Transformer decoder* [34] for the language model, which is a variant of the transformer [62]. This model applies a multi-headed self-attention operation over the input context tokens followed by position-wise feedforward layers to produce an output distribution over target tokens:

$$\begin{aligned} h_0 &= UW_e + W_p \\ h_l &= \texttt{transformer\_block}(h_{l-1}) \forall i \in [1, n] \\ P(u) &= \texttt{softmax}(h_n W_e^T) \end{aligned} \tag{2}$$

where $U = (u_{-k}, \ldots, u_{-1})$ is the context vector of tokens, $n$ is the number of layers, $W_e$ is the token embedding matrix, and $W_p$ is the position embedding matrix.

## 3.2 Supervised fine-tuning

After training the model with the objective in Eq. 1, we adapt the parameters to the supervised target task. We assume a labeled dataset $\mathcal{C}$, where each instance consists of a sequence of input tokens, $x^1, \ldots, x^m$, along with a label $y$. The inputs are passed through our pre-trained model to obtain the final transformer block's activation $h_l^m$, which is then fed into an added linear output layer with parameters $W_y$ to predict $y$:

$$P(y|x^1, \ldots, x^m) = \texttt{softmax}(h_l^m W_y). \tag{3}$$

This gives us the following objective to maximize:

$$L_2(\mathcal{C}) = \sum_{(x,y)} \log P(y|x^1, \ldots, x^m). \tag{4}$$

We additionally found that including language modeling as an auxiliary objective to the fine-tuning helped learning by (a) improving generalization of the supervised model, and (b) accelerating convergence. This is in line with prior work [50, 43], who also observed improved performance with such an auxiliary objective. Specifically, we optimize the following objective (with weight $\lambda$):

$$L_3(\mathcal{C}) = L_2(\mathcal{C}) + \lambda * L_1(\mathcal{C}) \tag{5}$$

Overall, the only extra parameters we require during fine-tuning are $W_y$, and embeddings for delimiter tokens (described below in Section 3.3).

这需要为每个单独的目标任务引入大量新参数，而我们在迁移过程中对模型架构的改动极小。

辅助训练目标

添加辅助的无监督训练目标是半监督学习的另一种形式。Collobert 和 Weston [10] 的早期工作使用了多种辅助 NLP 任务，如词性标注、组块分析、命名实体识别和语言建模，以改进语义角色标注。最近，Rei [50] 在其目标任务目标中添加了辅助语言建模目标，并在序列标注任务上展示了性能提升。我们的实验也使用了辅助目标，但正如我们所展示的，无监督预训练已经学习了与目标任务相关的多个语言方面。

## 3 框架

我们的训练过程包含两个阶段。第一阶段是在大规模文本语料库上学习一个高容量语言模型。随后是微调阶段，在此阶段我们使用带标签的数据将模型调整到判别性任务上。

### 3.1 无监督预训练

给定一个无监督的标记语料库 U = {u, . . . , u}，我们使用标准的语言建模目标来最大化以下似然函数：

$$L(U) = \sum_{i} \log P(u|u, \ldots, u; \Theta) \tag{1}$$

其中 k 是上下文窗口的大小，条件概率 P 通过参数为 Θ 的神经网络进行建模。这些参数使用随机梯度下降法 [51] 进行训练。

在我们的实验中，我们使用多层 Transformer 解码器[34]作为语言模型，该解码器是 Transformer[62]的一种变体。该模型对输入上下文标记应用多头自注意力操作，随后通过位置前馈层生成目标标记的输出分布：

$$\begin{aligned} h &= U W\_e + W\_p \\ h\_i &= transformer\_block(h\_\{i-1\})\ \forall i \in [1, n] \\ P(u) &= softmax(h\_n W\_e\text{\^{}}T) \end{aligned} \tag{2}$$

其中 $U = (u_1, \ldots, u_n)$ 是词元的上下文向量，n 是层数，$W_e$ 是词元嵌入矩阵，$W_p$ 是位置嵌入矩阵。

### 3.2 监督微调

在使用公式 1 中的目标训练模型后，我们将参数适配到监督目标任务。假设有一个带标签的数据集 C，其中每个实例由输入词元序列 $x^1, \ldots, x^m$ 以及标签 y 组成。输入通过预训练模型处理后，得到最终 Transformer 块的激活值 $h_l$，随后将其输入到一个新增的线性输出层（参数为 $W_y$）以预测 y：

$$P(y|x^1, \ldots, x^m) = softmax(h_l^m W_y) \tag{3}$$

由此我们得到以下需要最大化的目标函数：

$$L(C) = \sum_{(x,y)} \log P(y|x, \ldots, x)。 \tag{4}$$

我们还发现，将语言建模作为微调的辅助目标有助于学习，具体体现在：(a) 提升监督模型的泛化能力，(b) 加速收敛。这与先前的研究[50, 43]一致，这些研究也观察到此类辅助目标能带来性能提升。具体而言，我们优化以下目标函数（权重为λ）：

(5) 总体而言，微调过程中我们仅需额外引入的参数是 $W_y$ 以及分隔符标记的嵌入（详见下文第 3.3 节）。

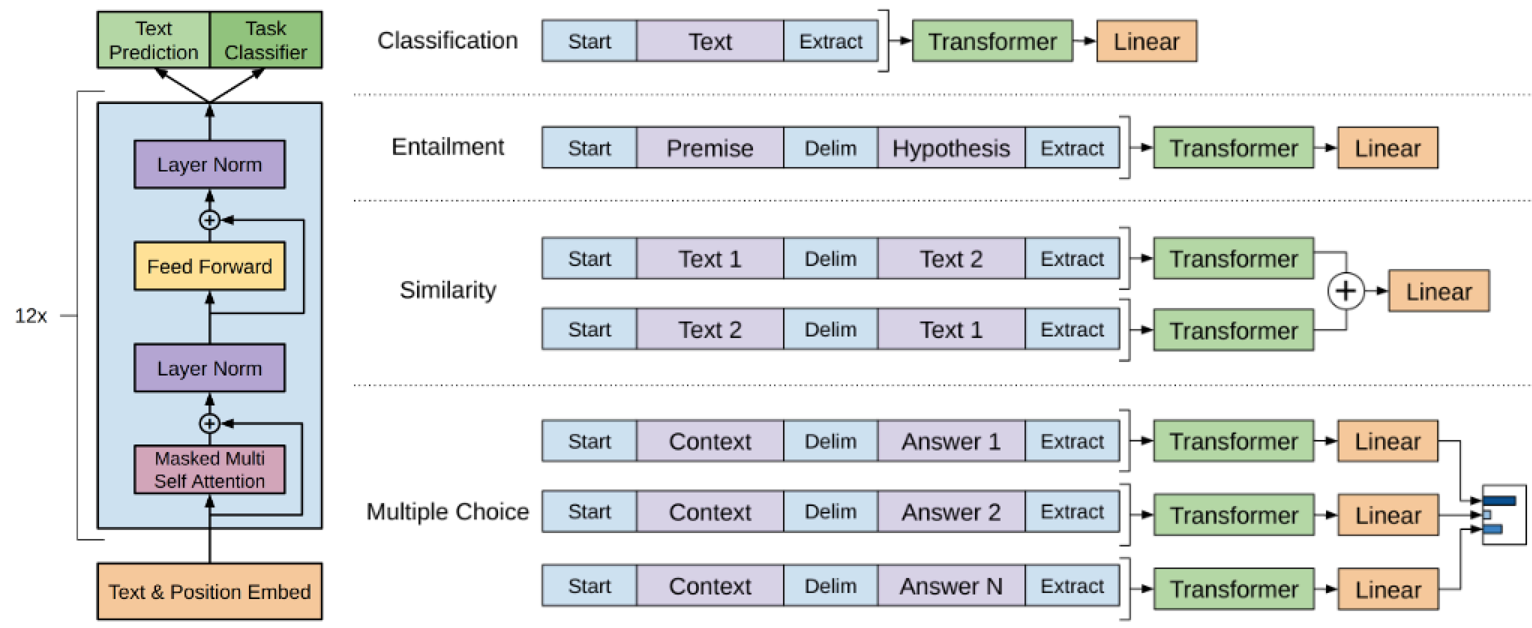


Figure 1: **(left)** Transformer architecture and training objectives used in this work. **(right)** Input transformations for fine-tuning on different tasks. We convert all structured inputs into token sequences to be processed by our pre-trained model, followed by a linear+softmax layer.

### 3.3 Task-specific input transformations

For some tasks, like text classification, we can directly fine-tune our model as described above. Certain other tasks, like question answering or textual entailment, have structured inputs such as ordered sentence pairs, or triplets of document, question, and answers. Since our pre-trained model was trained on contiguous sequences of text, we require some modifications to apply it to these tasks. Previous work proposed learning task specific architectures on top of transferred representations [44]. Such an approach re-introduces a significant amount of task-specific customization and does not use transfer learning for these additional architectural components. Instead, we use a traversal-style approach [52], where we convert structured inputs into an ordered sequence that our pre-trained model can process. These input transformations allow us to avoid making extensive changes to the architecture across tasks. We provide a brief description of these input transformations below and Figure 1 provides a visual illustration. All transformations include adding randomly initialized start and end tokens ($\langle s \rangle$, $\langle e \rangle$).

**Textual entailment** For entailment tasks, we concatenate the premise $p$ and hypothesis $h$ token sequences, with a delimiter token ($) in between.

**Similarity** For similarity tasks, there is no inherent ordering of the two sentences being compared. To reflect this, we modify the input sequence to contain both possible sentence orderings (with a delimiter in between) and process each independently to produce two sequence representations $h_l^m$ which are added element-wise before being fed into the linear output layer.

**Question Answering and Commonsense Reasoning** For these tasks, we are given a context document $z$, a question $q$, and a set of possible answers $\{a_k\}$. We concatenate the document context and question with each possible answer, adding a delimiter token in between to get $[z; q; \$; a_k]$. Each of these sequences are processed independently with our model and then normalized via a softmax layer to produce an output distribution over possible answers.

## 4 Experiments

### 4.1 Setup

**Unsupervised pre-training** We use the BooksCorpus dataset [71] for training the language model. It contains over 7,000 unique unpublished books from a variety of genres including Adventure, Fantasy, and Romance. Crucially, it contains long stretches of contiguous text, which allows the generative model to learn to condition on long-range information. An alternative dataset, the 1B Word Benchmark, which is used by a similar approach, ELMo [44], is approximately the same size

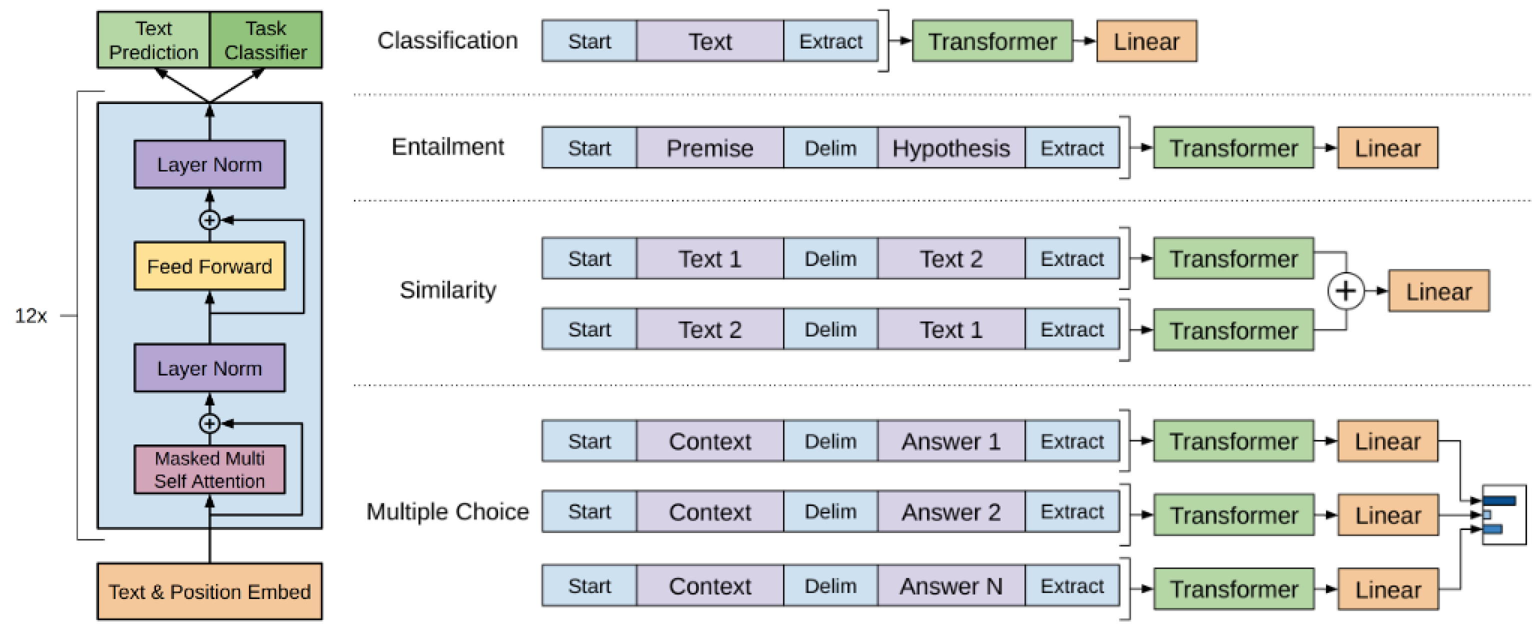


图 1:（左）本工作使用的 Transformer 架构及训练目标。（右）针对不同任务的输入变换。我们将所有结构化输入转换为标记序列，由预训练模型处理，随后接入线性层与 softmax 层。

### 3.3 任务特定的输入转换

对于某些任务，如文本分类，我们可以直接按照上述方法对模型进行微调。而其他任务，例如问答或文本蕴含，则具有结构化输入，如有序的句子对，或文档、问题和答案的三元组。由于我们的预训练模型是在连续的文本序列上训练的，因此需要对其进行一些修改才能应用于这些任务。先前的研究提出了在迁移表示的基础上学习任务特定的架构[44]。这种方法重新引入了大量任务特定的定制化工作，并且没有对这些额外的架构组件使用迁移学习。相反，我们采用了一种遍历式方法[52]，将结构化输入转换为预训练模型可以处理的有序序列。这些输入转换使我们能够避免在不同任务间对架构进行大规模修改。下面简要描述这些输入转换，图 1 提供了直观的说明。所有转换都包括添加随机初始化的起始和结束标记（⟨s⟩ ，⟨e⟩）。

文本蕴含 对于蕴含任务，我们将前提 p 和假设 h 的标记序列连接起来，中间用分隔符标记（$）隔开。

相似度 对于相似度任务，被比较的两个句子没有固有的顺序。为了反映这一点，我们修改输入序列，使其包含两种可能的句子顺序（中间用分隔符隔开），并分别独立处理每个序列，生成两个序列表示 h，在输入到线性输出层之前，将它们按元素相加。

问答与常识推理 对于这些任务，我们给定一个上下文文档 z、一个问题 q 和一组可能的答案{a}。我们将文档上下文和问题与每个可能的答案连接起来，中间添加一个分隔符标记，得到[z; q; $; a]。每个序列都使用我们的模型独立处理，然后通过 softmax 层进行归一化，以生成关于可能答案的输出分布。

## 4 实验

### 4.1 设置

无监督预训练 我们使用 BooksCorpus 数据集[71]来训练语言模型。该数据集包含超过 7000 本来自不同体裁（包括冒险、奇幻和浪漫）的未出版书籍。关键在于，它包含大量连续的文本段落，这使得生成模型能够学习依赖长距离信息。另一种类似方法 ELMo[44]所使用的替代数据集——1B 单词基准（1B Word Benchmark），其规模大致相当。

Table 1: A list of the different tasks and datasets used in our experiments.

| Task | Datasets |
|---|---|
| Natural language inference | SNLI [5], MultiNLI [66], Question NLI [64], RTE [4], SciTail [25] |
| Question Answering | RACE [30], Story Cloze [40] |
| Sentence similarity | MSR Paraphrase Corpus [14], Quora Question Pairs [9], STS Benchmark [6] |
| Classification | Stanford Sentiment Treebank-2 [54], CoLA [65] |

but is shuffled at a sentence level - destroying long-range structure. Our language model achieves a very low token level perplexity of 18.4 on this corpus.

**Model specifications** Our model largely follows the original transformer work [62]. We trained a 12-layer decoder-only transformer with masked self-attention heads (768 dimensional states and 12 attention heads). For the position-wise feed-forward networks, we used 3072 dimensional inner states. We used the Adam optimization scheme [27] with a max learning rate of 2.5e-4. The learning rate was increased linearly from zero over the first 2000 updates and annealed to 0 using a cosine schedule. We train for 100 epochs on minibatches of 64 randomly sampled, contiguous sequences of 512 tokens. Since layernorm [2] is used extensively throughout the model, a simple weight initialization of $N(0, 0.02)$ was sufficient. We used a bytepair encoding (BPE) vocabulary with 40,000 merges [53] and residual, embedding, and attention dropouts with a rate of 0.1 for regularization. We also employed a modified version of L2 regularization proposed in [37], with $w = 0.01$ on all non bias or gain weights. For the activation function, we used the Gaussian Error Linear Unit (GELU) [18]. We used learned position embeddings instead of the sinusoidal version proposed in the original work. We use the *ftfy* library[2] to clean the raw text in BooksCorpus, standardize some punctuation and whitespace, and use the *spaCy* tokenizer.[3]

**Fine-tuning details** Unless specified, we reuse the hyperparameter settings from unsupervised pre-training. We add dropout to the classifier with a rate of 0.1. For most tasks, we use a learning rate of 6.25e-5 and a batchsize of 32. Our model finetunes quickly and 3 epochs of training was sufficient for most cases. We use a linear learning rate decay schedule with warmup over 0.2% of training. $\lambda$ was set to 0.5.

### 4.2 Supervised fine-tuning

We perform experiments on a variety of supervised tasks including natural language inference, question answering, semantic similarity, and text classification. Some of these tasks are available as part of the recently released GLUE multi-task benchmark [64], which we make use of. Figure 1 provides an overview of all the tasks and datasets.

**Natural Language Inference** The task of natural language inference (NLI), also known as recognizing textual entailment, involves reading a pair of sentences and judging the relationship between them from one of `entailment`, `contradiction` or `neutral`. Although there has been a lot of recent interest [58, 35, 44], the task remains challenging due to the presence of a wide variety of phenomena like lexical entailment, coreference, and lexical and syntactic ambiguity. We evaluate on five datasets with diverse sources, including image captions (SNLI), transcribed speech, popular fiction, and government reports (MNLI), Wikipedia articles (QNLI), science exams (SciTail) or news articles (RTE).

Table 2 details various results on the different NLI tasks for our model and previous state-of-the-art approaches. Our method significantly outperforms the baselines on four of the five datasets, achieving absolute improvements of upto 1.5% on MNLI, 5% on SciTail, 5.8% on QNLI and 0.6% on SNLI over the previous best results. This demonstrates our model's ability to better reason over multiple sentences, and handle aspects of linguistic ambiguity. On RTE, one of the smaller datasets we evaluate on (2490 examples), we achieve an accuracy of 56%, which is below the 61.7% reported by a multi-task biLSTM model. Given the strong performance of our approach on larger NLI datasets, it is likely our model will benefit from multi-task training as well but we have not explored this currently.

[2] `https://ftfy.readthedocs.io/en/latest/`

[3] `https://spacy.io/`

表 1：我们实验中使用的不同任务和数据集列表。

| 任务 | 数据集 |
|---|---|
| 自然语言推理 | SNLI [5]、MultiNLI [66]、Question NLI [64]、RTE [4]、SciTail [25] |
| 问答 | RACE [30]、Story Cloze [40] |
| 句子相似度 | MSR Paraphrase Corpus [14]、Quora Question Pairs [9]、STS Benchmark [6] |
| 分类 | Stanford Sentiment Treebank-2 [54]、CoLA [65] |

ELMo [44] 在句子级别进行了打乱处理，从而破坏了长距离结构。我们的语言模型在该语料库上实现了极低的词元级困惑度，仅为 18.4。

模型规格 我们的模型主要遵循原始 Transformer 架构[62]。我们训练了一个 12 层仅解码器 Transformer，采用掩码自注意力头（768 维状态和 12 个注意力头）。对于逐位置前馈网络，我们使用了 3072 维内部状态。采用 Adam 优化方案[27]，最大学习率为 2.5e-4。学习率在前 2000 次更新中从零线性增加，随后通过余弦调度衰减至 0。我们在 64 个随机采样的连续 512 个 token 小批量上训练了 100 个周期。由于层归一化[2]在模型中广泛使用，简单的 N(0,0.02)权重初始化即可满足要求。我们使用包含 40000 次合并的字节对编码（BPE）词表[53]，并采用残差、嵌入和注意力 dropout（丢弃率 0.1）进行正则化。同时采用文献[37]提出的改进版 L2 正则化，对所有非偏置或增益权重设置 w=0.01。激活函数采用高斯误差线性单元（GELU）[18]。我们使用学习位置嵌入替代原始工作中提出的正弦位置编码。 我们使用 ftfy 库来清理 BooksCorpus 中的原始文本，标准化部分标点符号和空白，并使用 spaCy 分词器。

微调细节 除非另有说明，我们沿用无监督预训练中的超参数设置。我们在分类器中添加了丢弃率为 0.1 的 dropout。对于大多数任务，我们使用 6.25e-5 的学习率和 32 的批量大小。我们的模型微调速度很快，3 个训练周期在大多数情况下已足够。我们采用线性学习率衰减策略，并在训练的前 0.2% 进行预热。λ 设置为 0.5。

## 4.2 有监督微调

我们在多种有监督任务上进行了实验，包括自然语言推理、问答、语义相似度和文本分类。其中部分任务来自近期发布的 GLUE 多任务基准测试 [64]，我们对此进行了利用。图 1 展示了所有任务和数据集的概览。

自然语言推理 自然语言推理（NLI）任务，也称为文本蕴含识别，涉及阅读一对句子并从蕴含、矛盾或中立三种关系中判断它们之间的关系。尽管近期已有大量研究关注此任务[58, 35, 44]，但由于存在词汇蕴含、指代消解、词汇与句法歧义等多种语言现象，该任务仍具有挑战性。我们在五个来源多样的数据集上进行了评估，包括图像描述（SNLI）、转写语音、通俗小说与政府报告（MNLI）、维基百科文章（QNLI）、科学考试（SciTail）以及新闻文章（RTE）。

表 2 详细列出了我们的模型与先前最先进方法在不同 NLI 任务上的各项结果。我们的方法在五个数据集中的四个上显著优于基线模型，相较于此前最佳结果，在 MNLI 上实现了高达 1.5%的绝对提升，在 SciTail 上提升 5%，在 QNLI 上提升 5.8%，在 SNLI 上提升 0.6%。这证明了我们的模型在跨句子推理和处理语言歧义方面具有更强的能力。在 RTE（我们评估的较小数据集之一，含 2490 个样本）上，我们取得了 56%的准确率，低于多任务 biLSTM 模型报告的 61.7%。鉴于我们的方法在较大 NLI 数据集上的强劲表现，多任务训练很可能也会使我们的模型受益，但目前我们尚未对此进行探索。

[2] https://ftfy.readthedocs.io/en/latest/

[3] https://spacy.io/

Table 2: Experimental results on natural language inference tasks, comparing our model with current state-of-the-art methods. 5x indicates an ensemble of 5 models. All datasets use accuracy as the evaluation metric.

| Method | MNLI-m | MNLI-mm | SNLI | SciTail | QNLI | RTE |
|---|---|---|---|---|---|---|
| ESIM + ELMo [44] (5x) | - | - | <u>89.3</u> | - | - | - |
| CAFE [58] (5x) | 80.2 | 79.0 | <u>89.3</u> | - | - | - |
| Stochastic Answer Network [35] (3x) | <u>80.6</u> | <u>80.1</u> | - | - | - | - |
| CAFE [58] | 78.7 | 77.9 | 88.5 | <u>83.3</u> | | |
| GenSen [64] | 71.4 | 71.3 | - | - | <u>82.3</u> | 59.2 |
| Multi-task BiLSTM + Attn [64] | 72.2 | 72.1 | - | - | 82.1 | **61.7** |
| Finetuned Transformer LM (ours) | **82.1** | **81.4** | **89.9** | **88.3** | **88.1** | 56.0 |

Table 3: Results on question answering and commonsense reasoning, comparing our model with current state-of-the-art methods.. 9x means an ensemble of 9 models.

| Method | Story Cloze | RACE-m | RACE-h | RACE |
|---|---|---|---|---|
| val-LS-skip [55] | 76.5 | - | - | - |
| Hidden Coherence Model [7] | <u>77.6</u> | - | - | - |
| Dynamic Fusion Net [67] (9x) | - | 55.6 | 49.4 | 51.2 |
| BiAttention MRU [59] (9x) | - | <u>60.2</u> | <u>50.3</u> | <u>53.3</u> |
| Finetuned Transformer LM (ours) | **86.5** | **62.9** | **57.4** | **59.0** |

**Question answering and commonsense reasoning** Another task that requires aspects of single and multi-sentence reasoning is question answering. We use the recently released RACE dataset [30], consisting of English passages with associated questions from middle and high school exams. This corpus has been shown to contain more reasoning type questions that other datasets like CNN [19] or SQuaD [47], providing the perfect evaluation for our model which is trained to handle long-range contexts. In addition, we evaluate on the Story Cloze Test [40], which involves selecting the correct ending to multi-sentence stories from two options. On these tasks, our model again outperforms the previous best results by significant margins - up to 8.9% on Story Cloze, and 5.7% overall on RACE. This demonstrates the ability of our model to handle long-range contexts effectively.

**Semantic Similarity** Semantic similarity (or paraphrase detection) tasks involve predicting whether two sentences are semantically equivalent or not. The challenges lie in recognizing rephrasing of concepts, understanding negation, and handling syntactic ambiguity. We use three datasets for this task – the Microsoft Paraphrase corpus (MRPC) [14] (collected from news sources), the Quora Question Pairs (QQP) dataset [9], and the Semantic Textual Similarity benchmark (STS-B) [6]. We obtain state-of-the-art results on two of the three semantic similarity tasks (Table 4) with a 1 point absolute gain on STS-B. The performance delta on QQP is significant, with a 4.2% absolute improvement over Single-task BiLSTM + ELMo + Attn.

**Classification** Finally, we also evaluate on two different text classification tasks. The Corpus of Linguistic Acceptability (CoLA) [65] contains expert judgements on whether a sentence is grammatical or not, and tests the innate linguistic bias of trained models. The Stanford Sentiment Treebank (SST-2) [54], on the other hand, is a standard binary classification task. Our model obtains an score of 45.4 on CoLA, which is an especially big jump over the previous best result of 35.0, showcasing the innate linguistic bias learned by our model. The model also achieves 91.3% accuracy on SST-2, which is competitive with the state-of-the-art results. We also achieve an overall score of 72.8 on the GLUE benchmark, which is significantly better than the previous best of 68.9.

表 2：自然语言推理任务的实验结果，将我们的模型与当前最先进的方法进行了比较。5x 表示 5 个模型的集成。所有数据集均使用准确率作为评估指标。

| 方法 | MNLI-m MNLI-mm SNLI SciTail QNLI RTE |
|---|---|
| ESIM + ELMo [44] (5x) – – 89.3 – – CAFE [58] (5x) 80.2 79.0 89.3 – – 随机答案网络 [35] (3x) 80.6 80.1 – – – – | |
| CAFE [58] | 78.7 77.9 88.5 83.3 |
| GenSen [64] 71.4 71.3 – – 82.3 59.2 多任务 BiLSTM + 注意力机制 [64] 72.2 72.1 – – 82.1 61.7 微调后的 Transformer 语言模型（我们的方法）82.1 81.4 89.9 88.3 88.1 56.0 | |

表 3：在问答和常识推理任务上的结果，将我们的模型与当前最先进的方法进行比较。9x 表示 9 个模型的集成。

| 方法 | 故事完形填空 | RACE-m | RACE-h | RACE |
|---|---|---|---|---|
| val-LS-skip [55] | 76.5 | – | – | – |
| 隐式连贯性模型 [7] | 77.6 | – | – | – |
| 动态融合网络 [67] (9 倍) | – | 55.6 | 49.4 | 51.2 |
| 双向注意力 MRU[59]（9 倍） | – | 60.2 | 50.3 | 53.3 |
| 微调后的 Transformer 语言模型（我们的） | 86.5 | 62.9 | 57.4 | 59.0 |

问答与常识推理：另一项需要单一方面的任务
多句推理任务的一个典型例子是问答。我们采用了近期发布的 RACE 数据集[30]，该数据集包含来自中学和高中英语考试的段落及相关问题。研究表明，相较于 CNN[19]或 SQuAD[47]等其他数据集，该语料库包含更多需要推理能力的问题类型，这为我们的模型（专门训练用于处理长程上下文）提供了理想的评估场景。此外，我们还在故事完形测试[40]上进行了评估，该测试要求从两个选项中选择多句故事的正确结局。在这些任务中，我们的模型再次以显著优势超越了此前的最佳结果——在故事完形测试上提升达 8.9%，在 RACE 数据集上整体提升 5.7%。

这证明了我们的模型能够有效处理长距离上下文。

语义相似度 语义相似度（或释义检测）任务涉及预测两个句子是否语义等价。其挑战在于识别概念的改写、理解否定关系以及处理句法歧义。我们为此任务使用了三个数据集——微软释义语料库（MRPC）[14]（来自新闻源）、Quora 问题对数据集（QQP）[9]以及语义文本相似度基准（STS-B）[6]。我们在三个语义相似度任务中的两个上取得了最先进的结果（表 4），在 STS-B 上实现了 1 个百分点的绝对提升。在 QQP 上的性能差异显著，相比单任务 BiLSTM+ELMo+Attn 方法实现了 4.2%的绝对改进。

分类最后，我们还在两个不同的文本分类任务上进行了评估。语言可接受性语料库（CoLA）[65]包含专家对句子是否符合语法的判断，用于测试训练模型的内在语言偏向。而斯坦福情感树库（SST-2）[54]则是一个标准的二分类任务。我们的模型在 CoLA 上获得了 45.4 分，较此前最佳结果 35.0 分有显著提升，展现了模型学习到的内在语言偏向。该模型在 SST-2 上还达到了 91.3%的准确率，与当前最优结果相当。我们在 GLUE 基准测试中取得了 72.8 的总分，明显优于此前最佳成绩 68.9 分。

Table 4: Semantic similarity and classification results, comparing our model with current state-of-the-art methods. All task evaluations in this table were done using the GLUE benchmark. (*mc*= Mathews correlation, *acc*=Accuracy, *pc*=Pearson correlation)

| Method | Classification | | Semantic Similarity | | | GLUE |
|---|---|---|---|---|---|---|
| | CoLA (mc) | SST2 (acc) | MRPC (F1) | STSB (pc) | QQP (F1) | |
| Sparse byte mLSTM [16] | - | **93.2** | - | - | - | - |
| TF-KLD [23] | - | - | **86.0** | - | - | - |
| ECNU (mixed ensemble) [60] | - | - | - | 81.0 | - | - |
| Single-task BiLSTM + ELMo + Attn [64] | 35.0 | 90.2 | 80.2 | 55.5 | 66.1 | 64.8 |
| Multi-task BiLSTM + ELMo + Attn [64] | 18.9 | 91.6 | 83.5 | 72.8 | 63.3 | 68.9 |
| Finetuned Transformer LM (ours) | **45.4** | 91.3 | 82.3 | **82.0** | **70.3** | **72.8** |

Overall, our approach achieves new state-of-the-art results in 9 out of the 12 datasets we evaluate on, outperforming ensembles in many cases. Our results also indicate that our approach works well across datasets of different sizes, from smaller datasets such as STS-B (≈5.7k training examples) – to the largest one – SNLI (≈550k training examples).

## 5 Analysis

**Impact of number of layers transferred** We observed the impact of transferring a variable number of layers from unsupervised pre-training to the supervised target task. Figure 2(left) illustrates the performance of our approach on MultiNLI and RACE as a function of the number of layers transferred. We observe the standard result that transferring embeddings improves performance and that each transformer layer provides further benefits up to 9% for full transfer on MultiNLI. This indicates that each layer in the pre-trained model contains useful functionality for solving target tasks.

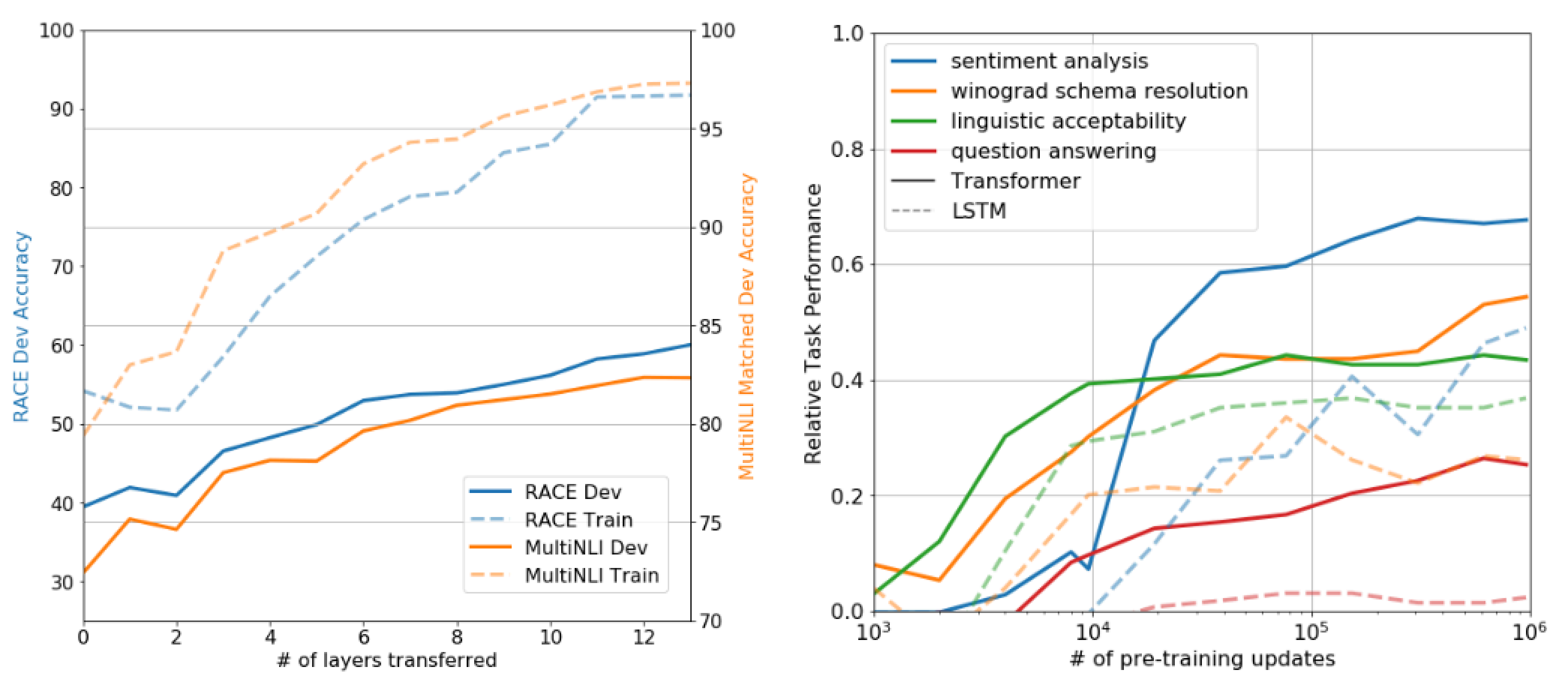


Figure 2: (**left**) Effect of transferring increasing number of layers from the pre-trained language model on RACE and MultiNLI. (**right**) Plot showing the evolution of zero-shot performance on different tasks as a function of LM pre-training updates. Performance per task is normalized between a random guess baseline and the current state-of-the-art with a single model.

**Zero-shot Behaviors** We'd like to better understand why language model pre-training of transformers is effective. A hypothesis is that the underlying generative model learns to perform many of the tasks we evaluate on in order to improve its language modeling capability and that the more structured

表 4：语义相似度与分类结果，将我们的模型与当前最先进方法进行对比。本表中所有任务评估均采用 GLUE 基准测试完成。（mc=马修斯相关系数，acc=准确率，pc=皮尔逊相关系数）

| 方法 | 分类 语义相似度 GLUE<br>CoLA SST2 MRPC STSB QQP (mc) (acc) (F1) (pc) (F1) |
|---|---|
| 稀疏字节 mLSTM [16] | – 93.2 – – – – |
| TF–KLD [23] | – – 86.0 – – – |
| ECNU（混合集成）[60] | – – – 81.0 – – |
| 单任务 BiLSTM + ELMo + 注意力机制 [64] 35.0 90.2 80.2 55.5 66.1 64.8 多任务 BiLSTM + ELMo + 注意力机制 [64] 18.9 91.6 83.5 72.8 63.3 68.9 | |
| 微调 Transformer 语言模型（我们的方法） | 45.4 91.3 82.3 82.0 70.3 72.8 |

总体而言，我们的方法在评估的 12 个数据集中有 9 个达到了新的最优结果，并在许多情况下超越了集成方法。我们的结果还表明，该方法在不同规模的数据集上均表现良好，从较小的数据集（如 STS–B，约 5700 个训练样本）到最大的数据集（如 SNLI，约 55 万个训练样本）。

## 5 分析

迁移层数的影响 我们观察了从无监督预训练到监督目标任务迁移不同层数的影响。图 2（左）展示了在 MultiNLI 和 RACE 数据集上，我们的方法性能随迁移层数变化的情况。我们观察到标准结果：迁移嵌入表示能提升性能，且每个 Transformer 层都能带来进一步收益，在 MultiNLI 上完全迁移时最高可提升 9%。这表明预训练模型中的每一层都包含解决目标任务的有用功能。

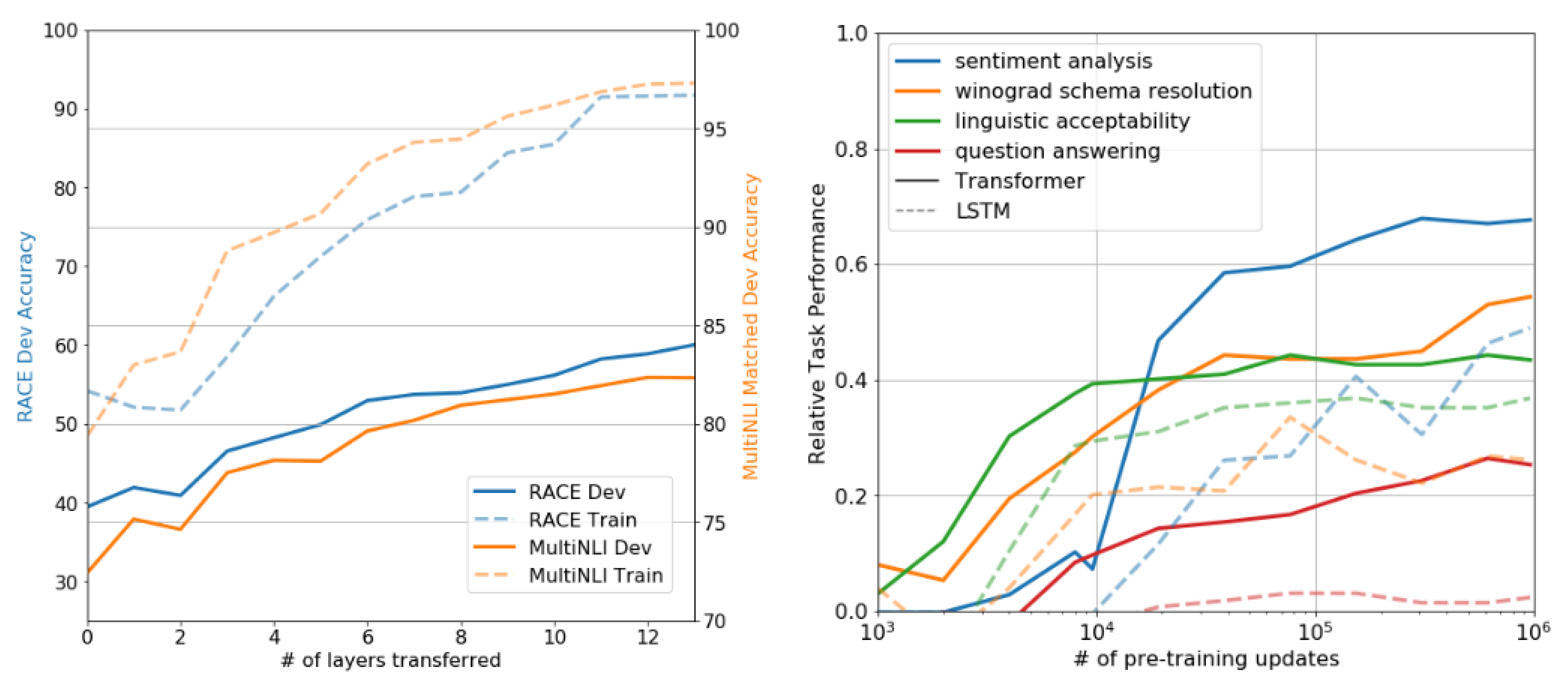


图 2：（左）从预训练语言模型迁移递增层数对 RACE 和 MultiNLI 的影响。（右）图表展示了不同任务上零样本性能随语言模型预训练更新次数的演变。每个任务的性能在随机猜测基线和当前单模型最优水平之间进行了归一化处理。

零样本行为 我们想更好地理解为什么 Transformer 的语言模型预训练如此有效。一个假设是，底层生成模型为了提升其语言建模能力，学会了执行我们评估的许多任务，并且结构越丰富

Table 5: Analysis of various model ablations on different tasks. Avg. score is a unweighted average of all the results. (*mc*= Mathews correlation, *acc*=Accuracy, *pc*=Pearson correlation)

| Method | Avg. Score | CoLA (mc) | SST2 (acc) | MRPC (F1) | STSB (pc) | QQP (F1) | MNLI (acc) | QNLI (acc) | RTE (acc) |
|---|---|---|---|---|---|---|---|---|---|
| Transformer w/ aux LM (full) | 74.7 | 45.4 | 91.3 | 82.3 | 82.0 | **70.3** | **81.8** | **88.1** | **56.0** |
| Transformer w/o pre-training | 59.9 | 18.9 | 84.0 | 79.4 | 30.9 | 65.5 | 75.7 | 71.2 | 53.8 |
| Transformer w/o aux LM | **75.0** | **47.9** | **92.0** | **84.9** | **83.2** | 69.8 | 81.1 | 86.9 | 54.4 |
| LSTM w/ aux LM | 69.1 | 30.3 | 90.5 | 83.2 | 71.8 | 68.1 | 73.7 | 81.1 | 54.6 |

attentional memory of the transformer assists in transfer compared to LSTMs. We designed a series of heuristic solutions that use the underlying generative model to perform tasks without supervised finetuning. We visualize the effectiveness of these heuristic solutions over the course of generative pre-training in Fig 2(right). We observe the performance of these heuristics is stable and steadily increases over training suggesting that generative pretraining supports the learning of a wide variety of task relevant functionality. We also observe the LSTM exhibits higher variance in its zero-shot performance suggesting that the inductive bias of the Transformer architecture assists in transfer.

For CoLA (linguistic acceptability), examples are scored as the average token log-probability the generative model assigns and predictions are made by thresholding. For SST-2 (sentiment analysis), we append the token *very* to each example and restrict the language model's output distribution to only the words *positive* and *negative* and guess the token it assigns higher probability to as the prediction. For RACE (question answering), we pick the answer the generative model assigns the highest average token log-probability when conditioned on the document and question. For DPRD [46] (winograd schemas), we replace the definite pronoun with the two possible referrents and predict the resolution that the generative model assigns higher average token log-probability to the rest of the sequence after the substitution.

**Ablation studies** We perform three different ablation studies (Table 5). First, we examine the performance of our method without the auxiliary LM objective during fine-tuning. We observe that the auxiliary objective helps on the NLI tasks and QQP. Overall, the trend suggests that larger datasets benefit from the auxiliary objective but smaller datasets do not. Second, we analyze the effect of the Transformer by comparing it with a single layer 2048 unit LSTM using the same framework. We observe a 5.6 average score drop when using the LSTM instead of the Transformer. The LSTM only outperforms the Transformer on one dataset – MRPC. Finally, we also compare with our transformer architecture directly trained on supervised target tasks, without pre-training. We observe that the lack of pre-training hurts performance across all the tasks, resulting in a 14.8% decrease compared to our full model.

## 6 Conclusion

We introduced a framework for achieving strong natural language understanding with a single task-agnostic model through generative pre-training and discriminative fine-tuning. By pre-training on a diverse corpus with long stretches of contiguous text our model acquires significant world knowledge and ability to process long-range dependencies which are then successfully transferred to solving discriminative tasks such as question answering, semantic similarity assessment, entailment determination, and text classification, improving the state of the art on 9 of the 12 datasets we study. Using unsupervised (pre-)training to boost performance on discriminative tasks has long been an important goal of Machine Learning research. Our work suggests that achieving significant performance gains is indeed possible, and offers hints as to what models (Transformers) and data sets (text with long range dependencies) work best with this approach. We hope that this will help enable new research into unsupervised learning, for both natural language understanding and other domains, further improving our understanding of how and when unsupervised learning works.

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表 5：不同任务上各类模型消融实验的分析结果。平均得分为所有结果的未加权平均值。（mc=马修斯相关系数，acc=准确率，pc=皮尔逊相关系数）

| 方法 | 平均得分 | CoLA (mc) | SST2 (acc) | MRPC (F1) | STSB (pc) | QQP (F1) | MNLI (acc) | QNLI (acc) | RTE (acc) |
|---|---|---|---|---|---|---|---|---|---|
| 带辅助语言模型的 Transformer（完整版） | 74.7 | 45.4 | 91.3 | 82.3 | 82.0 | 70.3 | 81.8 | 88.1 | 56.0 |
| 无预训练的 Transformer | 59.9 | 18.9 | 84.0 | 79.4 | 30.9 | 65.5 | 75.7 | 71.2 | 53.8 |
| 无辅助语言模型的 Transformer | 75.0 | 47.9 | 92.0 | 84.9 | 83.2 | 69.8 | 81.1 | 86.9 | 54.4 |
| 带辅助语言模型的 LSTM | 69.1 | 30.3 | 90.5 | 83.2 | 71.8 | 68.1 | 73.7 | 81.1 | 54.6 |

与 LSTM 相比，Transformer 的注意力记忆机制有助于迁移学习。我们设计了一系列启发式解决方案，利用底层生成模型在无需监督微调的情况下执行任务。我们在图 2（右）中展示了这些启发式方法在生成预训练过程中的有效性。我们观察到这些启发式方法的性能稳定且随训练过程持续提升，这表明生成式预训练支持学习多种任务相关功能。我们还注意到 LSTM 在零样本性能上表现出更高的方差，这暗示 Transformer 架构的归纳偏置有助于迁移学习。

对于 CoLA（语言可接受性任务），样本的评分依据生成模型分配的平均词元对数概率，并通过阈值设定进行预测。对于 SST–2（情感分析任务），我们在每个样本后附加词元"very"，并将语言模型的输出分布限制为仅包含"positive"和"negative"两个词，选择模型赋予更高概率的词元作为预测结果。对于 RACE（问答任务），我们基于文档和问题条件，选择生成模型赋予最高平均词元对数概率的答案。对于 DPRD [46]（维诺格拉德模式任务），我们将定指代词替换为两个可能的指代对象，并预测生成模型在替换后对序列剩余部分赋予更高平均词元对数概率的指代消解方案。

消融研究 我们进行了三项不同的消融研究（表 5）。首先，我们检验了在微调过程中不使用辅助语言模型目标时的方法性能。我们发现辅助目标对 NLI 任务和 QQP 有帮助。总体趋势表明，较大的数据集受益于辅助目标，而较小的数据集则不然。其次，我们通过将 Transformer 与使用相同框架的单层 2048 单元 LSTM 进行比较，分析了 Transformer 的效果。我们发现使用 LSTM 替代 Transformer 会导致平均得分下降 5.6 分。LSTM 仅在一个数据集（MRPC）上优于 Transformer。最后，我们还与直接在有监督目标任务上训练（无预训练）的 Transformer 架构进行了比较。我们发现缺乏预训练会损害所有任务的性能，与完整模型相比，性能下降了 14.8%。

## 6 结论

我们提出了一种框架，通过生成式预训练和判别式微调，利用单一任务无关模型实现强大的自然语言理解。通过在包含长距离连续文本的多样化语料库上进行预训练，我们的模型获得了丰富的世界知识和处理长距离依赖关系的能力，这些能力随后被成功迁移到解决判别式任务中，例如问答、语义相似度评估、蕴含关系判定和文本分类，在我们研究的 12 个数据集中有 9 个达到了最先进水平。利用无监督（预）训练来提升判别式任务的性能一直是机器学习研究的重要目标。我们的工作表明，实现显著的性能提升确实是可能的，并揭示了哪些模型（Transformer）和数据集（具有长距离依赖关系的文本）最适合这种方法。我们希望这能推动自然语言理解及其他领域无监督学习的新研究，进一步加深我们对无监督学习如何以及何时发挥作用的理解。

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